MAY-2026

R E S U M E

Full Name: Biniam Abrha Tsegay

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ACADEMIC DEGREES

2015- 2019: B.Sc. Civil and Environmental Engineering- Technion, Israel

2019 – 2021: M.Sc. Engineering and Management of Water Resources, Technion, Israel.
Thesis: "Storage water quality modeling - integration in water distribution systems management."
Supervisor:Professor Avi Ostfeld

2023-2027: PhD Researcher: Machine Learning & Intelligent Systems (School of Engineering, UBCO).

Supervisor: Dr. Nicolas Peleato.

PROJECTS

2020: Statistics Modeling Coronavirus expansion for Haifa city: implementing policies and simulating using Monto Carlo simulation.

2023 Determining the Chlorine dose using (UV-Vis) spectroscopy Data & supervised machine learning .

POSTER PRESENTATION AND SEMINARS.

Jan-2020 : "Storage water quality modeling - inclusion in water distribution systems management". Unity Day in the Environmental Engineering, Water and Agriculture Unit, Technion, Haifa, Israel, Prof. Avi Ostfeld

Dec-2020: "Storage water quality modeling - inclusion in water distribution systems management". GWRI- Water Research in Israel - the next generation of research, management, and industry.

Aug-2021: Seminar on “Analytical solutions to conservative and non-conservative water quality constituents in water distribution system storage tanks”, Technion, Haifa, Israel.

Jun -2024: “ Fuzzy Random Variable Analysis of Water Distribution System Demand”, CSCE Niagara 2024.

Nov-2024: Poster presentation, “Fuzzy Random Variable Analysis of Chlorine Residuals in Distribution Systems”, WQTC24 , Schaumburg, IL, USA.

May 30-2025 Generative Models for Data Augmentation in Smart Meter Water Consumption: A Conditional Approach, CSCE Winnipeg 2025

May 1-2026 "Improved Water Consumption Forecasting with Environment-Aware Generative AI"-BC Water & Waste Association (BCWWA) Conference, Penticton, Canada,

PUBLICATION

2021 **Biniam Abrha**; Avi Ostefld, Analytical Solutions to Conservative and Non-Conservative Water Quality Constituents in Water Distribution Systems Storage Tank, Water paper in MDPI. **https://doi.org/10.3390/w13243502**

2024 **Biniam Abrha**; Nicols M. Peleato,” Enhancing long-term water quality modeling by addressing base demand, demand patterns, and temperature uncertainty using unsupervised machine learning techniques” in Water Research. https://doi.org/10.1016/j.watres.2024.122701

2025 **Biniam Abrha**, Nicols M. Peleato,” Forecasting-Guided Generative AI for Improved Water Demand Forecasting Across Aggregation Scales” [Manuscript under review for publication]. Water Research

Conference Proceedings

2024 **Biniam Abrha,** Nicols M. Peleato,” Fuzzy Random Variable Analysis of Water Distribution System Demand” for the *Canadian Society for Civil Engineering (CSCE) 2024. https://link.springer.com/chapter/10.1007/978-3-031-97689-6_22*

2025 **Biniam Abrha,** Nicols M. Peleato,” Generative Models for Data Augmentation in Smart Meter Water Consumption: A Conditional

	Approach*”* In press, for the *Canadian Society for Civil Engineering (CSCE) 2025*.
2026	**Biniam Abrha**, Nicols M. Peleato. “*Integrating Environmental Drivers into Conditional Generative Models for Smart Water Demand Forecasting”*.

FELLOWSHIPS, AWARDS, AND HONORS.

2011-2015:	Full scholarship to study at Kellamino Special High School, a special boarding high school located in Tigray, Ethiopia. Only 90 students were accepted out of thousands of applicants based on a minimum threshold grade on the eighth Regional examination and passing a school-mandated admission exam.
2015:	Certificate of great distinction in the national exam of Ethiopia.
2015-2019:	Full scholarship to pursue a bachelor's degree in civil engineering at Technion in Israel. The initiative was made possible thanks to a partnership between the Gelfand Family Foundation, Tigray Development Association, and Technion. It was given to only 6 students out of 90 applicants, based on their cumulative high school grades (2011-2015) and Technion's oral entry tests.
2019-2021:	Marylin and Michael's winning fellowship, a full scholarship to complete an MSc degree in engineering and management of water resources at Technion.
2023-2027:	UBCO - Graduate Research Assistance(GRA) fellowship and International Doctorate Partial Tuition IDPT) Recipient.

BASIC SKILLS

Tools	Git/GitHub, Jupyter Notebook, VS Code, Google Colab, Linux command line, EPANET, WaterGEMS, ArcGIS, AutoCAD
Software:	Microsoft Word, Excel, PowerPoint
Programming	Python (NumPy, Pandas, TensorFlow, Keras, Scikit-learn, Matplotlib), R, MATLAB, LaTeX
Language Skills	English, Tigrigna, Amharic, Hebrew

Communication	Good communication/interpersonal skills gained through my role as a team member in community work in EWB, project presenter, and active participant in seminars and conferences.

WORK EXPERIENCE

2023-Present	Research Assistant
Fall 2024	Teaching Assistant ENGR_O 347 101: Environmental Engineering In this role, I grade assignments, Midterms, and final exams
Fall 2025	Teaching Assistant ENGR_O 347 101: Environmental Engineering ENGR_O 447 101:Desgin of process for water and wastewater treatment. In this role, I made tutorial videos, graded assignments, Midterms, and final exams

SPECIAL ACTIVITIES.

2016-2018:	Member of [Engineers without borders (EWB)] I was a dedicated volunteer for two years in EWB Technicon chapter. I was a member of the wind-group at the time. To provide electricity to a special needs school in east Jerusalem, our team designed and built a wind turbine from the bottom up. We also took part in educational programs promoting sustainability and green energy, as well as urging students to consider the environment and their carbon footprint.
2019-2022	Member of [Tigray innovative Inc] Starting from 2019, I am a member of innovative Tigray Inc, a nonprofit organization whose mission is to teach STEM (Science, Technology, Engineering, and Mathematics) online courses as well as assist Ethiopian high school students in obtaining scholarships to study abroad. I wrote syllabi and taught physics and mathematics.

REFERENCES

Prof. Avi Ostfeld
Professor at Technion-Israel Institute of Technology
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Personal website: https://cee.technion.ac.il/en/members/ostfeld/

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